PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Field	Details
Pentester Name	Pramita Shetty
Program / Batch	B082-Networkwalks Cyber security Internship
Date	August 21, 2026
Modules Completed	W2-PM1 (Multiple Kali Tools), W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks.com 2. Own local LAN network
Permission Secured?	Yes — authorized educational activity / own local network
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery
Phases 3–5	In Progress

1. Liability Disclaimer

I have performed these activities only on systems and networks where I had appropriate authorization or on devices/networks that I own. These materials are for educational and research purposes only. Reconnaissance and scanning must not be performed against unauthorized targets. Misuse of cybersecurity knowledge may result in legal, professional and financial consequences. All activities described in this report were completed within the scope of the assigned educational exercises.

2. Introduction

This report documents Week 2 cybersecurity practical activities covering two related penetration-testing phases: footprinting and reconnaissance of the networkwalks.com domain using multiple Kali Linux tools, and network discovery of a local LAN using Zenmap. Together, the activities demonstrate how a security professional can move from collecting publicly available information to identifying live hosts on an authorized network.

The footprinting activity used WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool provided a different view of the target, including registration information, web technologies, DNS resolution, HTTP headers, WAF identification and DNS records. The Zenmap activity used Windows networking commands and Zenmap's Ping Scan to discover active hosts and produce a network topology.

3. Objectives

- Understand the purpose of reconnaissance and footprinting in ethical hacking.
- Collect publicly available domain and web-technology information using Kali Linux tools.
- Resolve a domain to its IP address and inspect HTTP response headers.
- Identify whether a Web Application Firewall is present.
- Enumerate DNS records relevant to the target infrastructure.
- Identify the local IP address and subnet before performing network discovery.
- Discover live hosts and their IP/MAC addresses using Zenmap.
- Document technical findings, potential risks and practical recommendations.

4. Tools Used

Tool	Purpose
Kali Linux	Operating system used for reconnaissance activities.
WHOIS	Find domain registration details, dates and name servers.
WhatWeb	Fingerprint web technologies, server, CMS, plugins and related information.
Nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Inspect HTTP response headers, status and technical information.
Wafw00f	Detect whether a Web Application Firewall protects the website.
DNSRecon	Enumerate DNS records such as NS, MX, SPF, TXT and SRV records.
Windows CMD	Identify local IP, subnet and MAC address information.
Zenmap (Nmap GUI)	Discover live hosts and visualize the local network topology.

5. Activities Performed

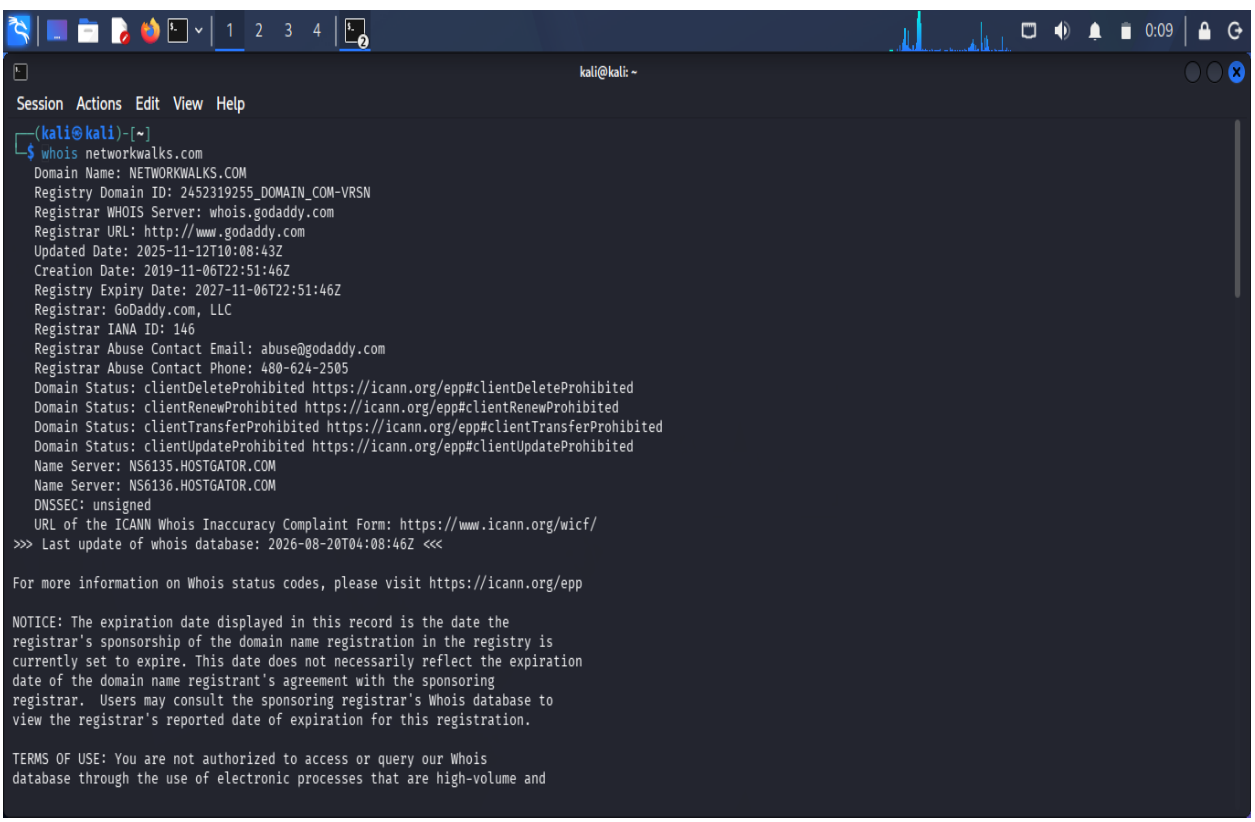
5.1 Footprinting & Reconnaissance

The reconnaissance activity was performed against the networkwalks.com domain as specified in the practical. Six Kali Linux tools were used. The objective was to build a profile of information that is already publicly exposed by the target.

Task 1 — WHOIS

$ whois networkwalks.com

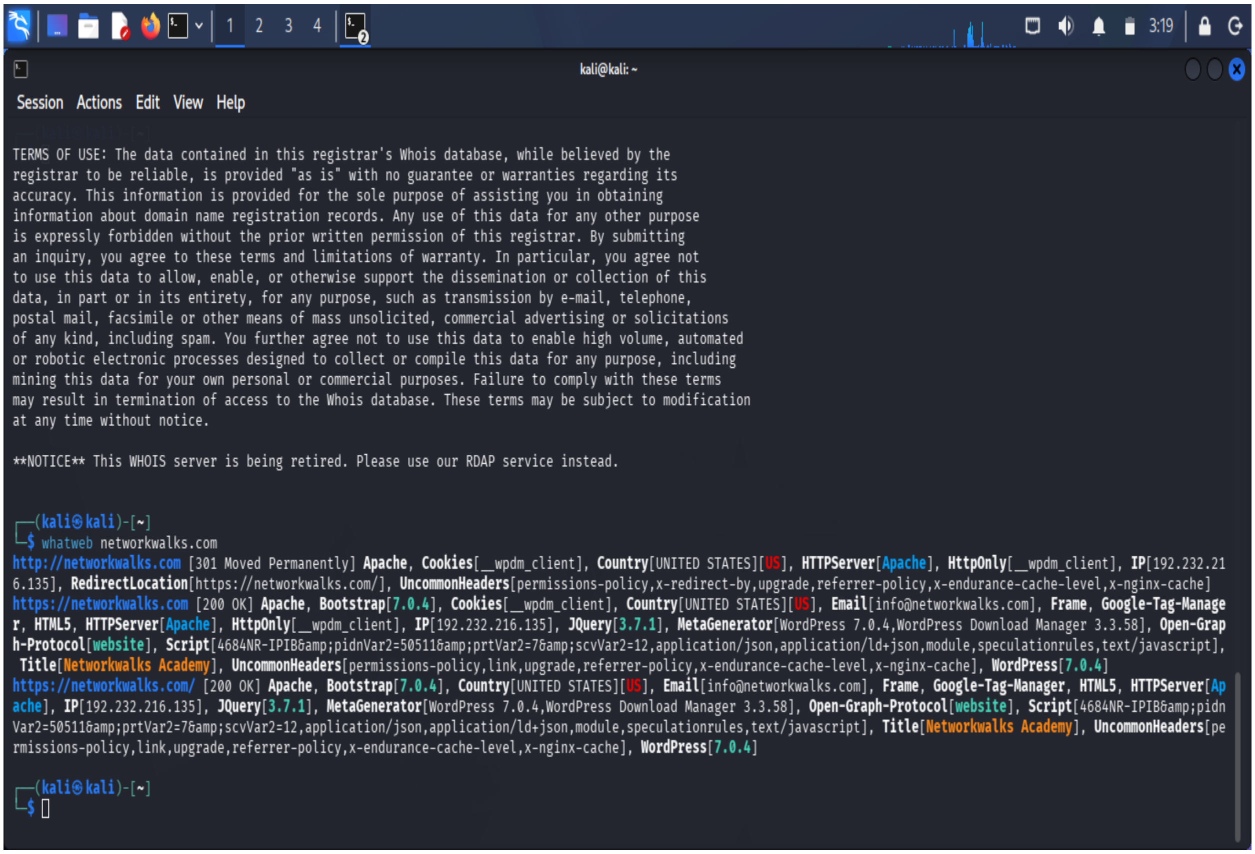
WHOIS was used to obtain public domain-registration information, including registrar/registration details and name servers. The practical notes that the name servers point to HostGator, providing an indication of the hosting provider.



Task 2 — WhatWeb

$ whatweb networkwalks.com

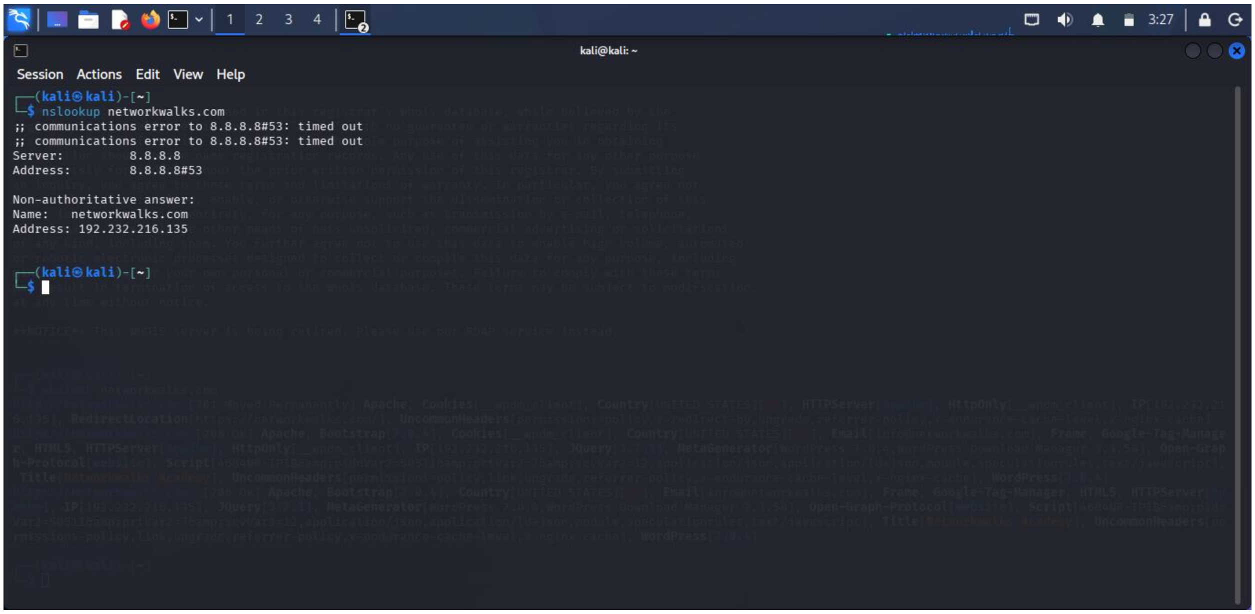
WhatWeb was used to fingerprint technologies exposed by the website. The practical identified WordPress 7.0.4 and WP Download Manager 3.3.58, together with additional web-server and contact information.



Task 3 — Nslookup

$ nslookup networkwalks.com

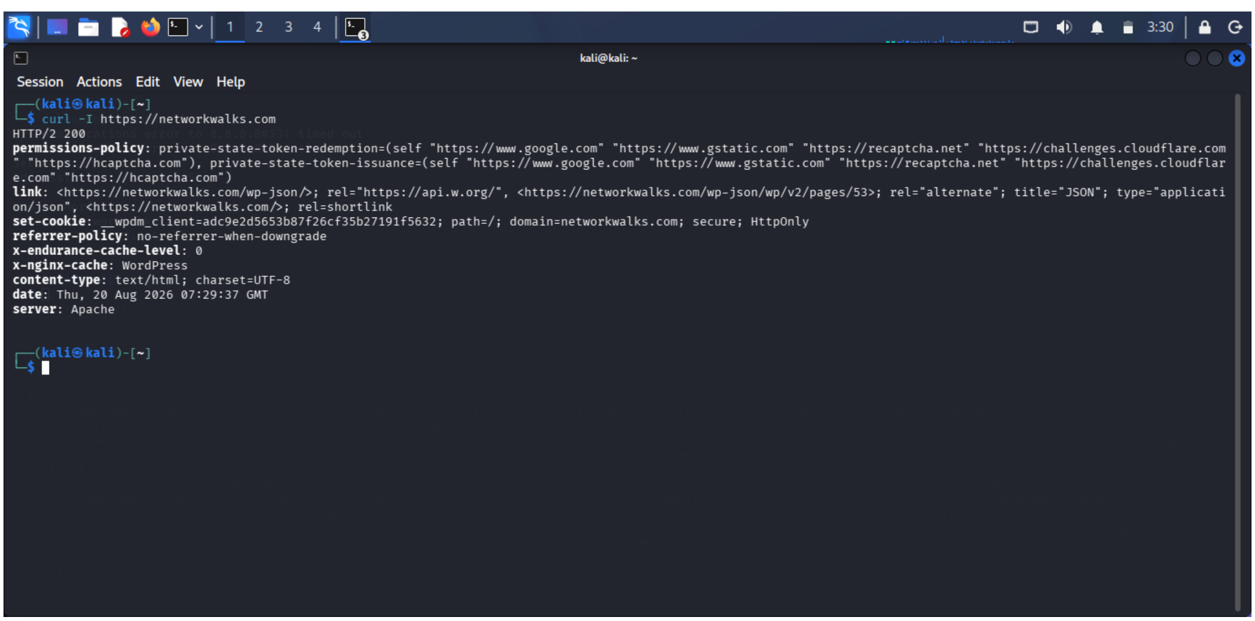
Nslookup resolved the domain name to the IP address 192.232.216.135 in the supplied practical result.



Task 4 — Curl

$ curl -I https://networkwalks.com

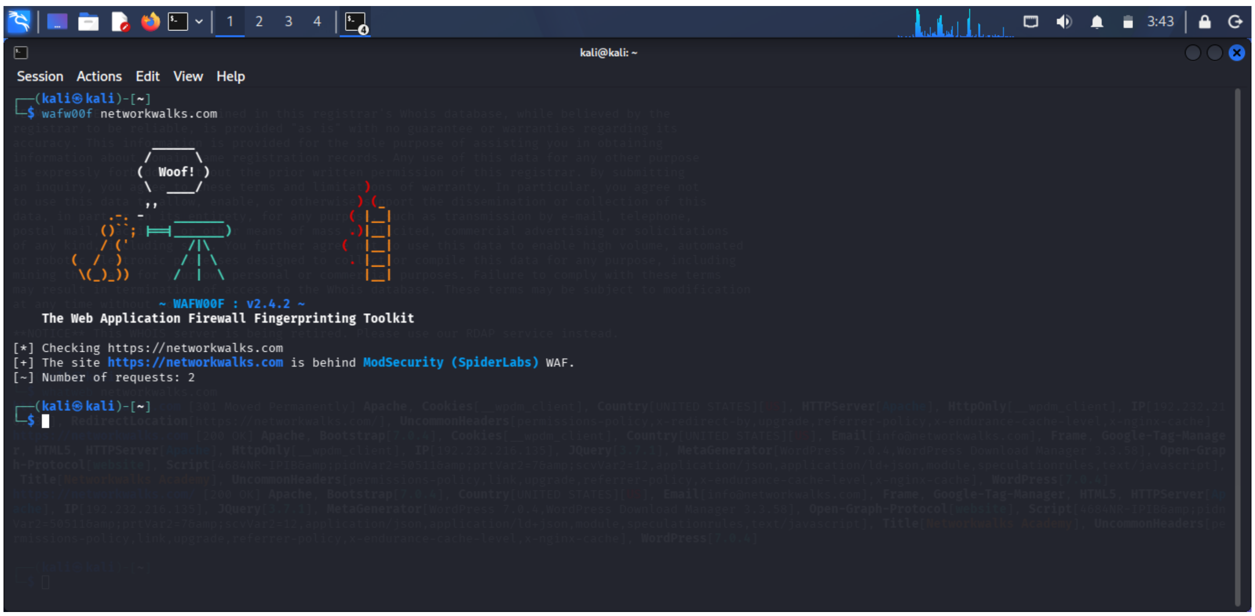
Curl was used to inspect HTTP response headers. The practical notes that the headers provided technical information and exposed the WordPress REST API endpoint /wp-json/.



Task 5 — Wafw00f

$ wafw00f networkwalks.com

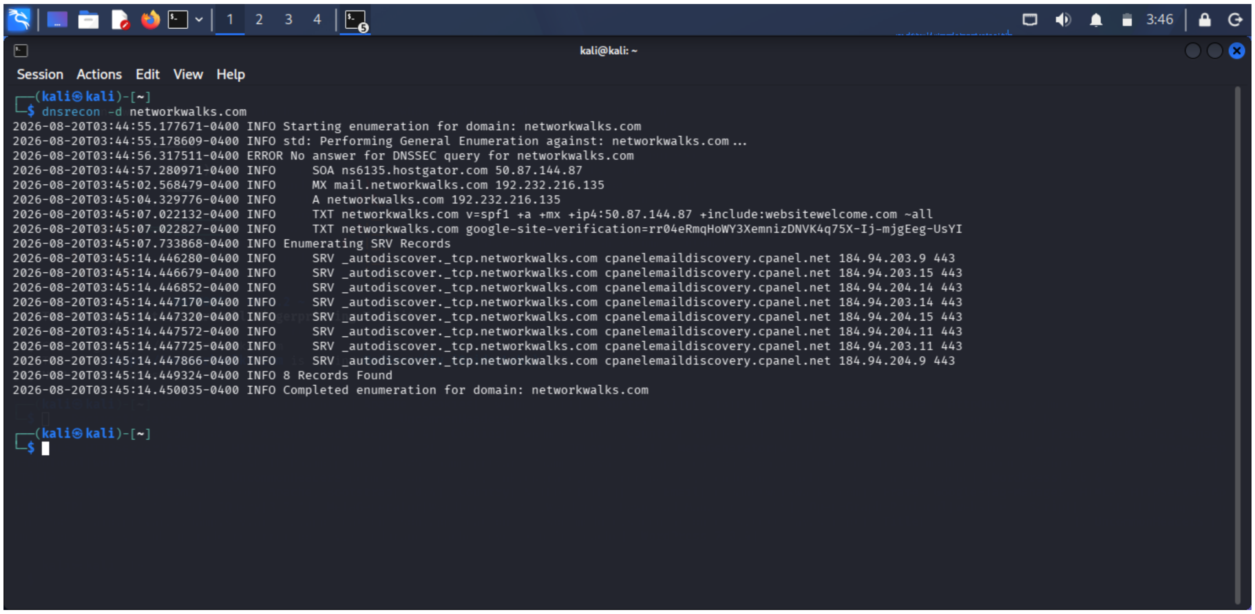
Wafw00f was used to detect a Web Application Firewall. The supplied result identified ModSecurity (SpiderLabs).



Task 6 — DNSRecon

$ dnsrecon -d networkwalks.com

DNSRecon was used to enumerate DNS information, including name servers, mail servers, SPF/TXT records, service records and DNS software information. The practical references Bind 9.16.23.



```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ dnsrecon -d networkwalks.com
2026-08-20T03:44:55.177671-0400 INFO Starting enumeration for domain: networkwalks.com
2026-08-20T03:44:55.178609-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-20T03:44:56.317511-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-20T03:44:57.280971-0400 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-08-20T03:45:02.568479-0400 INFO     MX mail.networkwalks.com 192.232.216.135
2026-08-20T03:45:04.329776-0400 INFO     A networkwalks.com 192.232.216.135
2026-08-20T03:45:07.022132-0400 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-20T03:45:07.022827-0400 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-08-20T03:45:07.733868-0400 INFO Enumerating SRV Records
2026-08-20T03:45:14.446280-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-08-20T03:45:14.446679-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-08-20T03:45:14.446852-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-08-20T03:45:14.447170-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-08-20T03:45:14.447320-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-08-20T03:45:14.447572-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-08-20T03:45:14.447725-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-08-20T03:45:14.447866-0400 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-08-20T03:45:14.449324-0400 INFO 8 Records Found
2026-08-20T03:45:14.450035-0400 INFO Completed enumeration for domain: networkwalks.com

┌──(kali㉿kali)-[~]
└─$
```

Phase 2

5.2 Network Scanning with Zenmap

The second practical focused on network discovery using Zenmap, the GUI version of Nmap. The exercise required identifying the local IP address and LAN subnet, scanning the subnet for live hosts, recording their IP and MAC addresses, and saving the resulting topology as a PDF.

1. Used the Windows ipconfig command to identify the local IP address and LAN subnet.
2. Entered the local subnet into Zenmap and selected Ping Scan to discover active hosts.
3. Recorded the number of live hosts and their IP addresses.
4. Collected MAC addresses for the discovered hosts.
5. Opened Zenmap's Topology view, enabled the legend and saved the topology as a PDF.

5.2.1 Supplied Practical Results

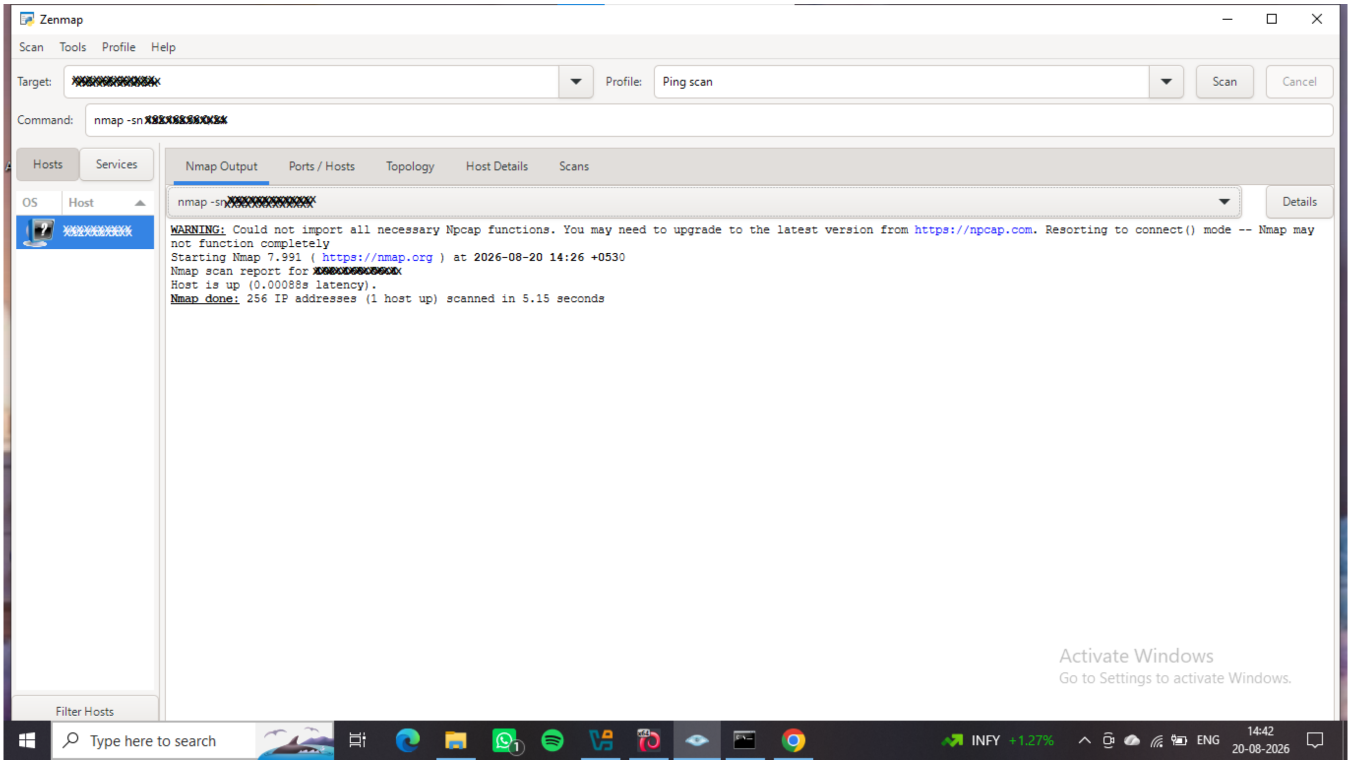
Item	Result from supplied practical
Live hosts	4 hosts (including the PC)
IP address 1	10.0.0.1
IP address 2	10.0.0.4
IP address 3	10.0.0.19
IP address 4	10.0.0.5
MAC address 1	00:50:56:E3:B3:2C
MAC address 2	00:0C:29:C0:94:8F
MAC address 3	00:50:56:E9:64:82
MAC address 4	00-0C-29-40-C0-93

Important: the Zenmap practical explicitly states that the actual subnet and results may differ depending on the local network configuration. Replace the supplied example values above with your own scan results before final submission if your instructor requires personal network evidence.

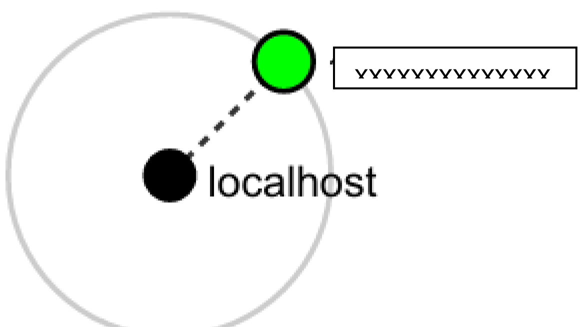
Note: IP address and MAC address here are just for example purpose.

```
C:\Users\HP>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : XXXXXXXXXXXXXXXXXXXXXXXXXXXXX
   IPv4 Address. . . . . . . . . . . : XXXXXXXXXXXXXX
   Subnet Mask . . . . . . . . . . . : XXXXXXXXXXXXXXXX
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . :XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
   IPv4 Address. . . . . . . . . . . : XXXXXXXXXXXXX
   Subnet Mask . . . . . . . . . . . : XXXXXXXXXXXXXX
   Default Gateway . . . . . . . . . : XXXXXXXXXXX

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

IP Config



Zenmap Scanning



Topology Result

6. Risk Analysis / Impact

The following items are observations from the footprinting and network-scanning exercises. They are not confirmed vulnerabilities. The activities primarily involved information gathering and host discovery; no exploitation or vulnerability validation was performed.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager.	Exposed technology/version details may help identify software requiring further security review.	Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135.	Provides information about the network location of the web service.	Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/.	May assist technology fingerprinting and further authorized enumeration.	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs).	Reveals information about the web application's security architecture.	Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records.	Can help build a broader infrastructure profile.	Medium
6	Multiple live hosts visible on local network	Zenmap identified four live hosts in the supplied practical example.	Unknown or unauthorized devices may potentially be present on a network.	Medium

Risk-level key: Critical | Medium | Low. The presence of a software version, IP address, DNS record or live host does not by itself prove that a system is vulnerable. Further authorized security testing would be required to validate any vulnerability.

7. Recommendations

1. Review publicly exposed technology information and minimize unnecessary disclosure of web-stack details.
2. Keep CMS platforms, plugins and other web technologies updated and review them against current security advisories.
3. Review HTTP response headers and reduce unnecessary technical information where appropriate.
4. Review DNS records regularly and ensure that only required information and services are publicly exposed.
5. Keep the identified WAF enabled, properly configured and monitored.
6. Perform regular internal network discovery to identify active and unexpected devices.
7. Investigate unknown devices found during network scans and verify that they are authorized.
8. Maintain current network topology and device documentation.
9. Perform reconnaissance and scanning only within an authorized scope.

8. Conclusion

During Week 2 of the Cybersecurity & Ethical Hacking practical program, I completed activities covering footprinting, reconnaissance and network scanning. In the footprinting activity, I used six Kali Linux tools to collect different categories of information about the target domain.

I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a Web Application Firewall, and DNSRecon can provide additional DNS information. These findings demonstrate how much information can be collected before any exploitation activity takes place.

In the network-scanning activity, I used Windows networking commands and Zenmap to identify local network configuration and discover active hosts. I also worked with IP and MAC address information and generated a network topology.

The exercises demonstrated that information gathering and documentation are important parts of cybersecurity. A security professional should understand what information is visible, assess what risks that information could create, and recommend controls to reduce unnecessary exposure. All reconnaissance and scanning should remain within an authorized scope.

9. Appendix — Commands Used

whois networkwalks.com

whatweb networkwalks.com

nslookup networkwalks.com

curl -I https://networkwalks.com

wafw00f networkwalks.com

dnsrecon -d networkwalks.com

ipconfig

ipconfig /all

Zenmap → Ping Scan → Local LAN subnet

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Project Information
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 |
Repository: https://github.com/pramita-06/networkwalks-B082-week2-Phase-2-3